Contents

1 Introduction

1.1 Motivation

The motivation should be highly relateable in the context of HCI.

- Challenges and existing problems in HCI
  - increasing cost
  - risk of mistakes
  - need of re-calibrating human scale
- Target domain
  - HCI sensors
  - using modern cg algorithms
- Scientific contribution
  - easy, intuitive, rapid method for prototyping sensors
  - perfect on body fit
  - flexible use i.e. positioning, scaling, copying, or reorienting a 2D pattern on the body can be a time consuming analog process.
  - potential to replicate, share with others, archive, fabricate remotely, or adapt to different bodies.
- Extendable domain of scanning other objects, plants, animals, food etc.
  - has wide application range
  - refer to Future Work section

1.2 Our approach

motivate our approach and describe in detail what it entails.
Using a handheld depth sensor camera, our digital tool provides guidance, feedback, and support of additional operations such as importing/exporting designs and 3d models.

1.3 Structure of the thesis

a small overview of what this thesis consists of in each chapter or rather what the reader can expect.

Chapter 2

Chapter 3

Chapter 4

Chapter 5

Chapter 6

2 Related Work

Our research builds upon existing work in gestural 3D modeling, skin-based fabrication, and fabrication-aware design.

Possible subsection: HCI work that clearly needs a method for rapid prototyping customized sensors.

2.1 Hybrid on-body fabrication

- ❑ what has been done
  - work that mainly focuses on using digital work flow to enhance prototyping process for on-body fabrication
  - Tactum, Exoskin, D-Coil, etc.
- ❑ their limitations and need for improvements
- ❑ strongly argue how i relate to them in HCI context
- ❑ research paper on review
  - Skinput: Appropriating the Body as an Input Surface → uses sensors for on-body transmission
  - Interactive Construction: Interactive Fabrication of Functional Mechanical Devices → analog design tool for precise fabrication
  - D-Coil: A Hands-on Approach to Digital 3D Models Design → hybrid fabrication
  - Interactive 3D garment design with constrained contour curves and style curves. Computer-Aided & DressUp: a 3D interface for clothing design with a physical mannequin → generalization of human anatomy for garment design. with our approach, we don't need any generalization.
  - Plushie: An Interactive Design System for Plush Toys → creating 2d cloth pattern and manipulating in 3D

2.2 Parameterization of depth cameras

- ❑ Literature review of other depth cameras (incl. technical background of how cameras perceive depth)
- ❑ limitations affecting the use in 3D scan domain
- ❑ Using Kinect v2 (features/liabilities and why we chose it)
- ❑ research paper on review
  - Using a Depth Camera as a Touch Sensor → depth camera primesense properties
  - A Data-Driven Approach for Real-Time Full Body Pose Reconstruction from a Depth Camera → using depth cameras in general

Kinect v2 has mainly been chosen because it provides a low-cost hand-held scanning alternative to us. Furthermore, it's sdk provides us with incredible correct measurement of real-world object scans to the digital work flow such that we can print 1:1 layers on the body.

Testing Kinect v2 dimensions trial 20 times and take average from it. real world measurement with help of third person. arm. 46 cm height and 10.5 cm width for arm. fingertip height was 8.5cm

50 cm was minimum distance to kinect before depth data appears distorted. everything beyond 100cm was perceived the same. moving in between 50cm and 100cm only yielded abyssal small differences.

0.5 - 4 meters for bodytracking from research we have:
+3m in z. -+0.2m in x. +-0.4m in y

Affecting Depth Sensors: reflective and light-absorbing materials(black) interferes with infrared light reflection back to camera. Infrared interference caused by two separated source might also distort depth data.
Kinect has usually no missing data but noisy/ (less) outliers data which makes surface reconstruction so hard.
dimensions:

	50 cm to kinect	100 cm to kinect
arm height. 46cm	45.18 cm	45.09 cm
arm width. 10.5cm	11.22 cm	10.52 cm
fingertip height 8.5cm	7.95 cm	7.87 cm

Observation: Every scan value was considered into the measurements which yielded a score of half the expected size. (for 10.5cm it wouldve been above 5cm). Because of the noises from kinect and lack of smoothing for joints detection, there sometimes were misfits in measured values .The dimensions perceived looked promising on consecutive perfect scans.

2.3 3D Scan applications

There are tons of other applications which says that they are scanning apps but only scans the camera view and creates a point cloud. they do not create a whole mesh. Those apps were excluded from the list. The following represents the best apps for 3D scanning from the kinect and reconstructing the surface to a water-tight mesh.

- Skanect only works on kinect v1 and others. it states that v2 does not provide good enough scanning results yields good result
- ReconstructMe only works on kinect v1 and others for example cant detect outliers of mouth. sometimes too superficial with reconstruction
- Fablitec only works on kinect v1 and others good results
- Recfusion only works on kinect v1 and others good results
- K-scan No longer supported. Cant be downloaded officially. no further information if kinect v2 is supported.
- 3D Scan app by microsoft very hard to yield good results (self-tested) microsoft advertises it but cant find other sources of how the reconstruction looks like
- Mobile apps.
  SCANN3D, Forge, Fabb-It, MobileFusion. But they mostly uses either their own cameras. Couldnt find one yet which utilizes the kinect.
- research paper on review
  –

From the research, the kinect v2 is not really popular among the competition of 3d scanning applications even though it has been proven that the version 2 has a way better depth perception than its predecessor. Only Skanect has stated that their compatibility with only the kinect 1 was due to its better performance to kinect 2.

2.4 Summary

- Lessons learned in regard to our approach

3 Methods

The theoretical part of my thesis. Here, i will talk about the mathematical work and why i decided on different algorithms to resolve the challenge of building a computational tool for rapid prototyping.

3.1 Pipeline

An overview pipeline of different modules interacting with each other. Goal of this section is to illustrate the interaction between the components.

1. Take volume from filter conditions
2. Use surface triangulation to create a near watertight mesh
3. Construct an acceleration structure
4. Extract Kinect camera orientation and map it to our virtual pinhole camera
5. Raycast and map each pixel point to an intersected smoothed triangle
6. Give this mapping as a framebuffer to the shader pipeline of KinectFusion $\Rightarrow$ receive and display shaded image buffer
7. Listen for mouse input and use its current position to apply mouse picking algorithm $\Rightarrow$ results in an intersected triangle
8. Annotate the triangle by storing the data
9. Build a graph G with the annotations
10. Unwrap algorithm on the graph $G \Rightarrow$ output svg

3.2 Kinect Fusion

- ❑ Introduction
  - system allows a user to pickup a standard Kinect camera and move rapidly within a room to reconstruct a high-quality, geometrically precise 3D model of the scene
  - avoids the reliance on RGB allowing use in in- door spaces with variable lighting conditions
  - reconstructing surfaces, which more accurately approximate real-world geometry
  - aim to perform camera track- ing without the need for prior augmentation of the space
- ❑ What do we use
- ❑ remark regarding restricted source code access
- ❑ tailored approach

3.3 Surface triangulation

Reconstructional method. We integrate this data using a volumetric representation[B. Curless and M. Levoy. A volumetric method for building complex models from range images.]

3.3.1 Filtering

- ❑ Conditional
- ❑ Outliers

3.3.2 ICP

using icp to align the virtual camera and to mesh the data to each other. Tracks the 6DOF pose of the camera and fuses live depth data from the camera into a single global 3D model in real-time.

3.4 Rendering Process

Generally, explain the necessity in rendering and how we use it under the points:

- ❑ Smoothed Triangle, intersection and shading geometry (might showcase difference to unsmoothed triangle)
- ❑ Super sampling (?)
- ❑ Using 3rd party shader (efficiency reasons)
- ❑ pseudo code

Illustrate rendering process theoretically.

3.4.1 BVH construction

- ❑ Parallelization
- ❑ Split-in-the-middle and SVH. tree depth termination criteria
- ❑ Bounding Box/ Slabs technique
- ❑ Tree traversal

3.4.2 Shading

- ❑ How does shading work and can't we just not use it? shader process explained

3.5 Mouse picking

concept of using raycast to get the triangle which the mouse points to. Used to annotate single triangles.

3.5.1 Arcball camera

- ❏ perspective camera implementation
- ❏ difficulties with the coordinate systems data from kinect

3.6 Unwrapping 3D volume

- ❏ why do we want this
- ❏ ongoing HCI challenges regarding this domain
- ❏ Math behind it with pseudocode

3.6.1 Integrating seams

How does using seams work with our algorithm.

4 Implementation

4.1 Work flow

- ❏ GUI
- ❏ Detailed step-by-step manual of how to use the application
- ❏ tailored to the methods chapter mentioned above

4.2 Software architecture

- ❏ c++, dependencies, concurrency and everything implemented
- ❏ image renderer
- ❏ synchronous work between models

5 Results

challenges and reason for the use case study.

5.1 Methodology

What kind of metrics do we use. how is it possible to evaluate intuitivness etc..

- ❏ Hypothesis
- ❏ Participants (classification)
- ❏ Procedure (questionnaire, unstructured interview)

5.2 Evaluation

- Concluding results
  - Try to resolve the hypothesis above, engage on the challenges proposed in the introduction. (flexibility, intuitiveness and rapidness)
- Prototyping real sensors
  - show the user the application range of the design tool. It is capable of creating a design for sensors.

6 Conclusion

6.1 Discussion

catch the main motivation from the introduction and try to mesh the results from the study into the outlook in regards to HCI. What was initially explored, how did we achieve it and how can we conclude the results.

6.2 Limitations

- Kinect and its sdk
  - has some flaws regarding infrared interferences
  - tons of noises that needs to be filtered correctly
- Our application
  - Unwrap algorithm is not yet capable to deal with more complex meshes
  - manipulation of camera angles sometimes feel clunky (system dependent)

6.3 Future work

- great to prototype sensors for any object shape i.e. on prosthetics
- can also be used to create individualistic 3d objects i.e. a tailor could digitally design a garment directly on a customers body, rather than manually measuring the body dimensions and subsequently producing the design.
- design with your fingers